

Chairperson: Voisin, Kevin

Secretary: Voisin, Kevin

Present:

Borgeat, Rémy

Carrino, Francesco

Voisin, Kevin

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Start of meeting

1 Administrative Organization

The meeting was held in person to establish the foundations of the collaboration.

- **Meeting Schedule:** Weekly meetings are planned for either Monday mornings or Fridays.
- **Responsibilities:** Mr. Voisin will plan and define the agenda (to be sent in advance if Mr. Carrino needs to review code or documents).
- **Deliverables:** Drafting of the Bachelor Thesis introductory report using the template: <https://github.com/hei-templates/hei-synd-prjspec/>
- Signature of the Bachelor Thesis form scheduled for May 15, 2026.

2 Technical Aspects and Data

2.1 Data Status (Constellium)

- **Volume:** Between 3 and 4 TB of data sampled at 40 kHz (ambient sound and accelerometer).
- **Signals:** Alloy type, tool paths (finishing vs. material removal), actual/setpoint speed, and torque.
- **Problem Statement:** Milling blades are currently changed too early (2-3 times per day). The goal is to extend their usage life by 2 to 3 times.

2.2 Storage

Data storage will initially be handled on an external SSD.

2.3 Previous Work

- 20 A correlation has been established between blade wear and torque. The current strategy is to predict torque using other sensor data.

3 Phases and Upcoming Deadlines

3.1 Week 1-2

- In-depth analysis and data visualization.
- 25 • State-of-the-art literature review.
- Production of the Project Specification document via the HEI template.

3.2 Immediate Calendar

- **Wednesday, 13.05.2026:** Data collection in room 23n313 (avoid the 10:00–11:00 slot).
- **Monday, 18.05.2026:** Submission of the initial document (Project Spec).
- 30 • **Action:** Creation of a private GitHub repository.

11:00 AM Meeting Adjourned